Does Showing Up Matter?
Lecture Attendance and Academic Outcomes in a University Course

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Abstract

We present an interactive visualization dashboard that explores the relationship between lecture attendance and academic performance across 15 years of student records from a university course at the University of West Bohemia. The dataset covers 2 740 evaluated students (academic years 2010/11–2024/25), each tracked over 13 lecture weeks per semester. Using a Plotly/Dash web application, we implemented seven coordinated views – including box plots, bar charts, scatter plots with an OLS trendline, and a Sankey flow diagram – backed by interactive filters for year, class group, teaching day, exam-attempt number, and attendance range. Our central finding is a steep pass-rate gradient: students attending 0–3 weeks pass at only 55 %, while those attending 10–13 weeks pass at 99 %. A moderate positive Pearson correlation ($r \approx 0.39$) between attendance and final points is confirmed across the full dataset. The COVID-19 remote-delivery period (AY 2019/20–2020/21) affected only the lowest-attendance cohort, whose pass rate fell by 17 percentage points, while higher-attendance groups were unaffected.

Keywords: information visualization, attendance analysis, academic performance, Plotly Dash, Sankey diagram

1. Introduction

Student absenteeism is a persistent concern in higher education. Lecturers, study advisors, and administrators routinely collect attendance records, yet these data are rarely analysed in a systematic and visually accessible way that would allow non-statistician stakeholders to draw actionable conclusions. The question of whether attendance actually predicts success – or merely correlates with latent motivation – has been studied in the educational psychology literature [1], but institution-level evidence over a long time horizon is scarce.

This work addresses that gap for one mandatory computer-science course at the University of West Bohemia. The dataset spans 15 academic years (2010/11 – 2024/25) and includes per-week binary attendance flags, final grades (1–4, S, N), final exam points, and the number of exam attempts for 2 740 evaluated students. Our goal is twofold: (1) to surface the empirical relationships between attendance and outcomes through a set of carefully chosen visualization types, and (2) to provide an interactive tool that lets instructors and study counsellors slice the data along dimensions that matter to them (year, class group, teaching day, etc.).

The primary research questions are: Does attendance predict final grade and exam points? Do low-attending students need more exam attempts to pass? Is there an attendance threshold above which pass rates become near-universal? What does the full attendance → attempts → grade pathway look like in aggregate?

2. Related Work

Meta-analyses of class attendance in tertiary education find consistent positive correlations between attendance and grades [1]. Effect sizes vary across disciplines and institutions, but the relationship is sufficiently robust to justify attendance-based early-warning interventions.

On the visualization side, educational data mining commonly uses bar charts and scatter plots to communicate grade distributions and correlations [2]. Sankey diagrams have been applied to student-flow analyses, tracing cohorts through enrolment, retention, and graduation stages [3]. Interactive dashboards built with web-based frameworks expose complex multi-dimensional data to users who do not have a programming or statistics background; Plotly Dash [4] is a widely used

Python framework for this purpose. OLS regression trendlines embedded in scatter plots are a standard technique for making linear associations visible without requiring the reader to interpret a correlation coefficient [5].

3. Methods

3.1. Data and preprocessing

The dataset consists of one row per student per academic year with the following columns: `academic_year`, `class`, `day` (teaching day of the week), per-week binary attendance flags `week_1` through `week_13`, `total_attendance` (sum of attended weeks, 0–13), `exam_attempt` (integer, missing when no exam was taken), `final_grade` (1, 2, 3, 4, S, or N), and `final_points` (numeric, available for a subset of years).

Preprocessing is performed in pandas [6]: weekly flags are summed to produce `total_attendance`; students are classified into four attendance brackets (0–3, 4–6, 7–9, 10–13 attended weeks); and each student is assigned an outcome label (*Passed 1st attempt*, *Passed (multiple attempts)*, or *Did not pass*) based on whether any of their records contain a passing grade (1, 2, 3, 4, or S) and on the minimum exam-attempt number at which that grade was achieved.

3.2. Visualization dashboard

The application is built with Python, Plotly Express, and Dash [4]. It exposes seven view types selectable from a dropdown:

1. **Attendance vs Final Grade** – box, line, or scatter plot showing the distribution of attendance per grade category.
2. **Attendance Distribution** – box, line, or scatter plot showing how many students attended each number of lectures.
3. **Exam Attempts vs Attendance** – box, line, or scatter plot relating the number of exam attempts to total attendance.
4. **Pass Rate by Attendance Bracket** – grouped bar or line chart showing the fraction of students who passed, needed multiple attempts, or did not pass within each bracket.
5. **Final Points vs Attendance** – scatter plot with an OLS trendline; box and line variants also available.
6. **Attendance vs Grade Flow (Sankey)** – a Sankey diagram tracing students from attendance bracket through exam-attempt number to final grade.
7. **Attendance by Teaching Day** – box plot comparing attendance distributions across Monday–Friday cohorts.

Each view can be further restricted by five interactive filters: academic year, class group, teaching day, exam-attempt number, and an attendance range slider (0–13). Four KPI cards at the top of the dashboard always reflect the currently visible cohort: student count, average attendance, exam-participation rate, and the number of displayed rows out of the total.

When a user switches view, the dashboard automatically snaps the chart type and filter defaults to the most informative preset for that view (e.g., the Sankey view defaults to the largest class group; the Points vs Attendance view restricts to the most recent year and students who have a recorded final-points value).

3.3. OLS trendline

For the scatter-plot view of final points vs attendance, a custom OLS trendline is computed with `numpy.polyfit` (degree 1) and rendered as a dashed black overlay. The legend reports the fitted slope and R^2 so that the reader can assess the strength of the linear relationship at a glance.

4. Results

4.1. Pass rate by attendance bracket

The starkest finding is the step-wise increase in pass rate across the four attendance brackets (Fig. 1). Students attending 0–3 lectures pass at only 55 %, while the rate rises to 91 % for 4–6 weeks, 95 % for 7–9 weeks, and 99 % for 10–13 weeks. The jump between the lowest and the next bracket (+36 pp) is by far the largest, suggesting that crossing the threshold of attending roughly one-third of lectures is the single most important behavioural factor for passing.

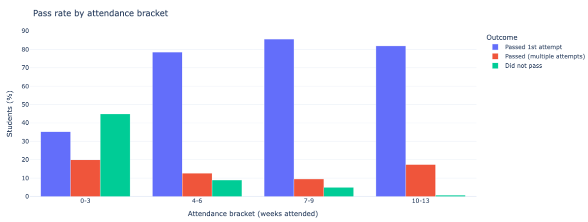

Figure 1: Pass rate by attendance bracket. The grouped bar chart separates *Passed 1st attempt*, *Passed (multiple attempts)*, and *Did not pass* for each of the four brackets.

4.2. Attendance and final points

The scatter plot with OLS trendline (Fig. 2) shows a moderate positive linear relationship between total attendance (0–13 weeks) and final exam points. The Pearson correlation across the full dataset is $r \approx 0.39$. The fitted slope indicates that each additional attended lecture is associated with roughly 1.5–2 additional points on the final exam. Substantial scatter around the trendline confirms that attendance is a contributing factor, not a deterministic one: a sizeable minority of high-attenders score low, and some low-attenders score high.

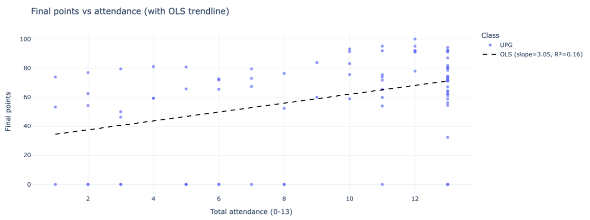

Figure 2: Scatter plot of final points vs total attendance with OLS trendline ($r \approx 0.39$). Each point represents one student; colour encodes class group.

4.3. Attendance and final grade

The box plot in Fig. 3 shows the distribution of attendance for each final grade category. Students who received the best grade (1) attended on average more than 3 additional lectures compared with students who failed (N or S). The medians decrease monotonically from grade 1 to N, and the inter-quartile ranges overlap substantially, again consistent with a moderate rather than deterministic relationship.

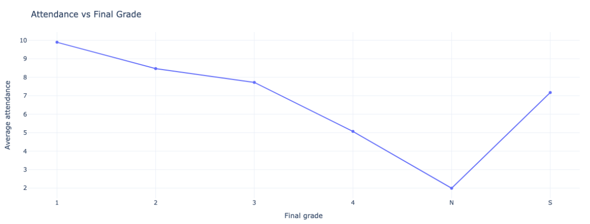

Figure 3: Attendance distribution per final grade. Grade 1 is the best outcome; N indicates failure.

4.4. Exam attempts and attendance

Fig. **??** shows that students who needed more exam attempts had, on average, lower attendance. First-attempt passers cluster near the upper end of the attendance range, whereas repeat-attempt students show a notably lower median and more low-attendance outliers.

4.5. Sankey flow diagram

The Sankey diagram (Fig. 4) traces the complete pathway from attendance bracket through exam-attempt number to final grade. It makes visually apparent that the failure stream – students ending up with grade N – is almost entirely fed by the 0–3 bracket, while the 10–13 bracket overwhelmingly flows into first-attempt grade-1 or grade-2 outcomes. The diagram also reveals that a non-trivial flow from the 4–6 bracket requires two or three attempts before passing.

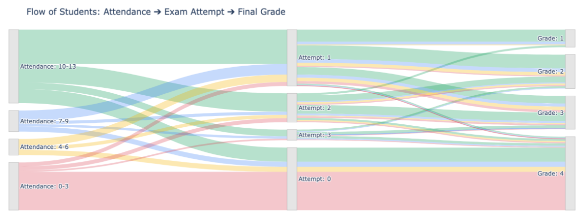

Figure 4: Sankey diagram: attendance bracket → exam-attempt number → final grade. Link colour is inherited from the source attendance bracket.

4.6. COVID-19 remote-delivery period

Restricting the data to AY 2019/20 and 2020/21 (352 students, average attendance 6.09/13) reveals that the pandemic affected outcomes asymmetrically. Low-attenders (0–3 weeks) saw their pass rate fall from 55 % to 38 % (-17 pp). Medium-attenders (4–9 weeks) were stable or marginally improved (≈93 % → ≈96 %). High-attenders (10–13 weeks) were virtually unaffected (99 % → 98 %). This finding suggests that online delivery disproportionately hurt students who were already the most disengaged.

5. Discussion

The positive association between attendance and performance is stable across 15 academic years and robust to the COVID disruption at moderate and high attendance levels. However, the moderate correlation ($r \approx 0.39$) means that attendance explains roughly 15 % of the variance in final points. The remaining 85 % is driven by factors outside this dataset: prior knowledge, study habits, ability, and motivation.

The most important caveat is **self-selection bias**. Students who are motivated and well-prepared tend both to attend more often and to perform better. The association we observe is therefore not necessarily causal. A randomised experiment assigning compulsory vs optional attendance would be needed to isolate the causal effect.

A second limitation is that the dataset covers a single course at a single institution, which limits generalisability. The attendance-performance gradient might be weaker in courses with different pedagogical structures (e.g., fully recorded lectures or problem-based learning).

Third, students who withdrew from the course or never registered for the exam are excluded from the pass-rate calculations, which may slightly inflate the apparent pass rate for low-attendance students (those students may have been even worse performers who are not counted as failures).

Despite these limitations, the dashboard is practically useful. The steep pass-rate jump at the 4-week threshold provides a clear, actionable boundary for early-warning systems: students absent in the first three weeks of the semester can be flagged for outreach before the midterm, while there is still time to improve attendance.

6. Future Work

Several extensions would deepen the analysis. First, examining per-week attendance curves (weeks 1–13 individually) could reveal whether early-semester absence is a stronger predictor than late-semester absence, enabling finer-grained intervention timing.

Second, integrating a simple logistic-regression or gradient-boosting classifier into the dashboard would allow instructors to obtain, after the first few weeks, a probability estimate for each student of eventually passing the course.

Third, expanding the dataset to multiple courses across the faculty would enable cross-subject comparisons and test whether the attendance–performance gradient generalises beyond a single course.

Fourth, adding anonymised prior-achievement data (e.g., high-school leaving-exam scores) would make it possible to partial out the self-selection effect via regression adjustment or propensity-score matching, yielding a cleaner causal estimate.

Finally, the dashboard could be extended with a real-time data-import pipeline so that lecturers can monitor the current semester's attendance in the same tool, rather than relying on retrospective analysis.

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References

[1] M. Credé, S. G. Roch, U. M. Kieszczynka, Class attendance in college: A meta-analytic review of the relationship of class attendance with grades and student characteristics, Review of Educational Research 80 (2010) 272–295. doi:10.3102/0034654310362998.

[2] C. Romero, S. Ventura, Educational data mining: A review of the state of the art, IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews) 40 (2010) 601–618. doi:10.1109/TSMCC.2010.2053532.

[3] L. Aulck, N. Velagapudi, J. Blumenstock, J. West, Predicting student dropout in higher education, in: Proceedings of the ICML Workshop on #Data4Good: Machine Learning in Social Good Applications, 2017.

[4] Plotly Technologies Inc., Collaborative data science, 2015. Montréal, QC. Available at `https://plotly.com`.

[5] W. S. Cleveland, The Elements of Graphing Data, Wadsworth Advanced Books and Software, Monterey, CA, 1985.

[6] W. McKinney, Data structures for statistical computing in Python, in: S. van der Walt, J. Millman (Eds.), Proceedings of the 9th Python in Science Conference, 2010, pp. 56–61.